



1. Consider a system where the page fault service time is 200 ms, and the main memory access time is 10 ms. The translation look aside buffer is added to improve the performance. The 80% reference are found in the TLB and that of the remaining 10% cause page faults. The TLB access time is 2 ms and the page table access time is negligible. Then what is the effective memory access time (EMAT)?

(a) 15.8ms (b) 12.8ms
(c) 13.8ms (d) 16.8 ms

2. If the disk head is located initially at 32, find the number of disk moves required with FCFS if the disk queue of I/O blocks requests are 98, 37, 14, 124, 65, 100.

(a) 354 (b) 239
(c) 322 (d) 200

3. An operating system uses the Banker's algorithm for deadlock avoidance when managing the allocation of three resource types X, Y and Z to three processes P0, P1 and P2. The table given below presents the current system state. Here, the Allocation matrix shows the current number of resources of each type allocated to each process and the Max matrix shows the maximum number of resources of each type required by each process during its execution.

	Allocation			Max		
	X	Y	Z	X	Y	Z
P0	0	0	1	8	4	3
P1	3	2	0	6	2	0
P2	2	1	1	3	3	3

There are 3 units of type X, 2 units of type Y and 2 units of type Z still available. The system is currently in a safe state. Consider the following independent requests for additional resources in the current state:

REQ1: P0 requests 0 units of X, 0 units of Y and 2 units of Z.

REQ2: P1 requests 2 units of X, 0 units of Y and 0 units of Z

Which one of the following is TRUE?

- Only REQ1 can be permitted
- Only REQ2 can be permitted.
- Both REQ1 and REQ2 can be permitted
- Neither REQ1 nor REQ2 can be permitted.

4. Consider the following implementation as a solution to the critical section problem for two processes P₀ and P₁ with id 0 and 1, respectively.

```
int flag [2] = {0, 0}
int turn = 0 :
void lock (int id) /*id = 0 or 1*/
{
    turn = id ^ 1; /*S1*/
    flag [id] = 1; /*S2*/
}
```

```
/* '^' is the bitwise XOR*/
while (flag [id ^ 1]) && turn == (id ^ 1):
}
```

```
void unlock (int id)
```

```
{
    flag [id] = 0;
}
```

which of the following statements are correct?

- Solution leads to a deadlock
- Mutual exclusion is guaranteed if S1 and S2 are interchanged
- Mutual exclusion is guaranteed
- None of the above

5. Consider a disk pack with 32 surfaces, 128 tracks per surface, and 256 sectors per track. 512 bytes of data are stored in a bit-serial manner in a sector. The capacity of the disk pack and the number of bits required to address a particular sector in the disk are respectively.

- 1024 MB, 21
- 128 MB, 18
- 512 MB, 20
- 256 MB, 20

6. Consider the 3 processes, P1, P2 and P3 shown in the table

Process	Arrival Time	Time unit Required
P1	0	5
P2	1	7
P3	3	4

The completion order of the 3 processes under the policies FCFS and RR2 (round robin scheduling with CPU quantum of 2 time units) are

- FCFS : P1, P2, P3, RR2: P1, P2, P
- FCFS : P1, P3, P2 RR2: P1, P3, P2
- FCFS : P1, P2, P3, RR2: P1, P3, P2
- FCFS : P1, P3, P2 RR2: P1, P2, P3

7. One disk queue with requests for I/O to blocks on cylinders. The Requests are in the following manner. 98, 183 37, 122 14 124 65 67 considering SSTF (shortest seek time first)scheduling ,the total number of head movements is ,if the disk head of initially at 53 is :

(a) 236 (b) 246
(c) 220 (d) 240

8. Match the following

List-I	List-II
I. Stack Algorithm	P. Deadlock
II. Elevator algorithm	Q. Disk Scheduling
III. Priority scheduling Algorithm	R. Page Replacement
IV. Havender's Algorithm	S. CPU Scheduling



1. I-R, II-Q, III-S, IV-P 2. I-Q, II-R, III-S, IV-P
3. I-R, II-Q, III-P, IV-S 4. I-Q, II-R, III-P, IV-S

9. Consider the following two threads T1 and T2 that update two shared variables a and b. Assume that initially $a = b = 1$. Though context switching between threads can happen at any time, each statement of T1 and T2 is executed atomically, without interruption.

T1 T2
 $a = a + 1$ $b = 2 + b$
 $b = b + 1$ $a = 2 + a$

Which one of the following options lists all the possible combinations of values of a and b after both T1 and T2 finish execution.

- (a) $(a=4, b=4)$ $(a=3, b=3)$ $(a=4, b=3)$
(b) $(a=3, b=4)$ $(a=4, b=3)$ $(a=3, b=3)$
(c) $(a=4, b=4)$ $(a=4, b=3)$ $(a=3, b=4)$
(d) $(a=2, b=2)$ $(a=2, b=3)$ $(a=3, b=4)$

10. Consider the following threads, T_1, T_2, T_3 executing on a single processor, synchronized using three binary semaphore variables, S_1, S_2 and S_3 operated upon using standard wait () and signal (). The thread can be context switched in any order and at any time.

T_1	T_2	T_3
while (true) { wait (S_3) ; print ("C") ; signal (S_2) ; }	while {true} { wait (S_1) ; print ("B") ; signal (S_3) ; }	while (true) { wait (S_2) ; print ("A") ; signal (S_1) ; }

- (a) $S_1 = 1, S_2 = 1, S_3 = 1$
(b) $S_1 = 1, S_2 = 1, S_3 = 0$
(c) $S_1 = 1, S_2 = 0, S_3 = 0$
(d) $S_1 = 0, S_2 = 1, S_3 = 1$

11. Determine the number of page faults when references to pages occur in the order 1,2,4,5,2,1,2,4. Assume that the main memory can accommodate 3 pages and the main memory already has the pages 1 and 2, with page 1 bought earlier than page 2. (Assume LRU) i.e. Least-Recently-Used algorithm is applied
- (a) 3 (b) 4
(c) 5 (d) None of the above

12. Consider the following segment table in segmentation scheme

Segment ID	Base	Limit
0	200	200
1	5000	1210
2	1527	498
3	2500	50

What happens if the logical address requested is - Segment ID 2 and offset 1000?

- A. Fetches the entry at the physical address 2527 for segment ID 2.
B. A trap is generated
C. Deadlock
D. Fetches the entry at offset 27 Segment ID 3.

13. In a particular Unix OS, each data block is of size 1024 bytes, each node has 10 direct data block addresses and three additional addresses : one for single indirect block, one for double indirect block and one for triple indirect block. Also, each block can contain addresses for 128 blocks. Which one of the following is approximately the maximum size of a file system?

- (a) 512 MB (b) 2 GB
(c) 8 GB (d) 16 GB

14. Suppose the time to service a page fault is on the average 10 milliseconds, while a memory access takes microsecond. Then a 99.99% hit ratio results in average memory access time of

- (a) 1.9999 milliseconds
(b) 1 millisecond
(c) 9.999 microseconds
(d) 1.9999 microseconds

15. Consider a system having 'm' resources of the same type. the resources are shared by 3 processes A,B,C which have peak time demands of 3,4,6 respectively. The minimum value of 'm' that ensures that deadlock will never occur is

- (a) 11 (b) 12
(c) 13 (d) 14

16. Consider the below program as two concurrent processes

Semaphore x : = 0

P1: repeat forever P2: repeat forever

1. V(x); 4. P(x);
2. compute; 5. Compute;
3. P(x); 6. V (x) ;

Consider the following statement about process P1 & P2:

- (1) It is possible for process P1 to starve.
(2) It is possible for process P2 to starve.
Which of the following is true?

- (a) Both 1 & 2 are true
(b) Both 1 & 2 are false
(c) 1 is true & 2 is false
(d) 2 is true & 1 is false

17. Consider a 32-bit system with 4 KB page size and page table entries of size 4 bytes each. Assume 1 KB = 2^{10} bytes. The OS uses a 2-level page table for memory management, with the page table containing an outer





page directory and an inner page table. The OS allocates a page for the outer page directory upon process creation. The OS uses demand paging when allocating memory for the inner page table, i.e. a page of the inner page table is allocated only if it contains at least one valid page table entry.

An active process in this system accesses 2000 unique pages during its execution, and none of the pages are swapped out to disk. After it completes the page accesses, let X denote the minimum and Y denote the maximum number of pages across the two levels of the page table of the process.

The value of $X + Y$ is _____

18. A process has 5 pages (0 to 4) and a reference string, 0,1,2,3,0,1,4,0,1,2,3,4

A 3- frame FIFO page replacement policy is used. How many page faults will occur?

Given options:

- (a) 9 (b) 10
(c) 11 (d) 12

19. A system has 3 processes (P1, P2, P3) and 3 resource types (A,B,C)

Available Resources:

A = 3, B = 3, C = 2

Allocation Matrix

Process	A	B	C
P1	0	1	0
P2	2	0	0
P3	3	0	3

Max Need Matrix

Process	A	B	C
P1	7	5	3
P2	3	2	2
P3	9	0	3

Is the system in a safe state?

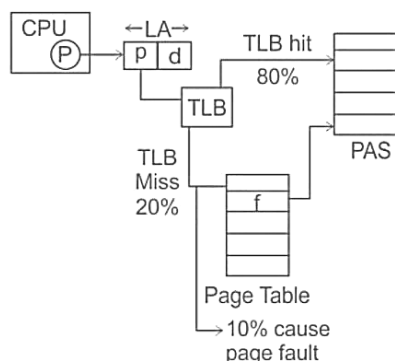
option

- (a) Yes, safe state
(b) No, deadlock
(c) Need more information
(d) Cannot be determined
20. Disk requests : 82,170,43,140,24,16,190
Initial head position : 50
Options:
(a) 208 cylinders
(b) 214 cylinders
(c) 196 cylinders
(d) 225 cylinders



SOLUTION

1.



Data:

TLB hit ratio = 80%, TLB miss ratio = 20%

TLB access time = 2 ms

Main memory access time = 10 ms

Page fault rate = 10%, page miss rate = 90%

Page fault service time = 200 ms

Calculation:

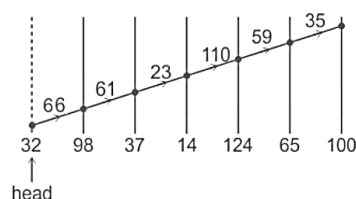
EMAT = TLB hit ratio (TLB time + main memory time) + TLB miss ratio [TLB time + page fault rate (page fault service time) + page miss rate (main memory time)]

EMAT = 0.8 (2+10) + 0.2 [2 + 0.1 (200) + 0.9 (10)]

EMAT = 9.6 + 6.2

EMAT = 15.8 ms

2. Solution:



Calculation:

Total Head movement = (98-32) + (98-37) + (37-14) + (124-14) + (124-65) + (100-65)

Total Head movement to serve all request

= 66 + 61 + 23 + 110 + 59 + 35 = 354

3. Solution.

Allocation				Max			Need		
	X	Y	Z	X	Y	Z	X	Y	Z
P0	0	0	1	8	4	3	8	4	2
P1	3	2	0	6	2	0	3	0	0
P2	2	1	1	3	3	1	2	2	2
Available X = 3, Y = 2, Z = 2									

REQ1 asks X = 0, Y = 0, Z = 2 for P0

Now, If the request REQ1 is permitted, the state would become

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	Allocation			Max			Need		
	X	Y	Z	X	Y	Z	X	Y	Z
P0	0	0	3	8	4	3	8	4	2
P1	3	2	0	6	2	0	3	0	0
P2	2	1	1	3	3	1	2	2	2
Available X = 3, Y = 2, Z = 0									

Notice that in table P0's allocated Z becomes 3 from 1 and Available resources of Z reduce to 0 from 2.

with the above table we can use available resources to service P1.

	Allocation			Max			Need		
	X	Y	Z	X	Y	Z	X	Y	Z
P0	0	0	3	8	4	3	8	4	2
P1	3	2	0	6	2	0	0	0	0
P2	2	1	1	3	3	3	1	2	2
Available X = 3, Y = 2, Z = 0 becomes Available X = 6, Y = 4, Z = 0									

with 0 instances of Z being available, no requests of P0 or P2 can be fulfilled, hence the system is in deadlock.

Therefore, REQ1 cannot be permitted.

Therefore REQ1 cannot be permitted

REQ2 asks X = 2, Y = 0, Z = 0 for P1

Now, if the request REQ2 is permitted, the state would become.

Allocation				Max		Need			
	X	Y	Z	X	Y	Z	X	Y	Z
P0	0	0	3	8	4	3	8	4	2
P1	5	2	0	6	2	0	1	0	0
P2	2	1	1	3	3	1	2	2	2
Available X = 1, Y = 2, Z = 2									





Notice that in table P1's allocated X becomes 5 from 3 and Available resources of X reduce to 1 from 3.
with the above table, we can use available resources to service P1, (P2 can also be serviced)

	Allocation			Max			Need		
	X	Y	Z	X	Y	Z	X	Y	Z
P0	0	0	1	8	4	3	8	4	2
P1	5	2	0	6	2	0	0	0	0
P2	2	1	1	3	3	3	1	2	2

Available X = 1, Y = 2, Z = 2 becomes Available X = 6, Y = 4, Z = 2

With the above available instances, P2 can be serviced and then P0.

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Final table will look like,

	Allocation			Max			Need		
	X	Y	Z	X	Y	Z	X	Y	Z
P0	0	0	1	8	4	3	0	0	0
P1	5	2	0	6	2	0	0	0	0
P2	2	1	1	3	3	3	0	0	0

Available X = 1, Y = 2, Z = 2 becomes Available X = 6, Y = 4, Z = 2

Therefore, REQ1 can be permitted.

4. Solution:

Mutual exclusion:

Mutual exclusion is a process synchronization characteristic that "no two processes may exist in the crucial region at any one moment."

Deadlock:

A deadlock occurs when two computer programs that share the same resources effectively prevent each other from accessing the resource, allowing both processes to stop working. The first computer operating systems could only run one application at a time.

Step 1:

Initially Here, /*id = 0 or 1*/ and S1 and S2 are

Step 2:

turn = id ^ 1 ; / *S1*/ , turn = 1 (if id = 0 or 1)

Step 3:

flag [id] = 1; / *S2*/ flag = 1

Step 4:

while (flag [id ^ 1]) && turn == (id ^ 1):

If the process p1 starts the execution, the condition also flag [1] && turn == 1 written true and starts the next statement after that the process P1 is preempted and P2 starts execution this is also entering the execution of remaining statements it means two processes may exist in the crucial region at any one moment.

Hence mutual exclusion is guaranteed if S1 and S2 are interchanged.

5. Solution:

32 surfaces = 5 bits

128 tracks = 7 bits

256 sectors = 8 bits

sector size 512 bytes = 9 bits

The capacity of disk = $2^{5+7+8+9} = 2^{29} = 512 \text{ MB}$

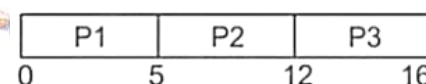
To specify a particular sector we do not need sector size, so bits required,

= 5 + 7 + 8 = 20

Thus, the Capacity of the disk pack and the number of bits required is 512 MB, 20 respectively.

6. Solution:

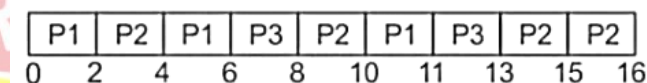
The GANTT chart for the FCFS scheduling algorithm is



The completion order for the FCFS scheduling algorithm is :

P1 P2 P3

The GANTT chart for the RR scheduling algorithm is



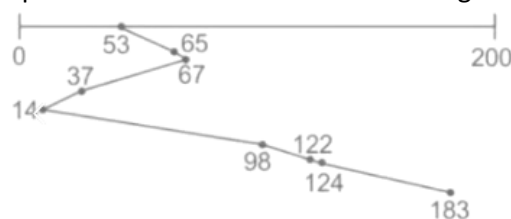
The completion order for the RR scheduling algorithm is : P1

P3 P2

hence, the correct answer is "option 3".

7. Solution:

For example, after servicing block 53. the nearest block request is at 65 and moves to block 65. Similarly all above requests are serviced as shown in the figure below.



Total number of head movements = (67-53) + (67-14) + (183-14) = 236

8. Option(A)

1. I-R, II-Q, III-S, IV-P





9. Solution:

1. $a = a + 1$
2. $b = b + 1$
3. $b = 2 * b$
4. $a = 2 * a$

The processor has 4 lines to execute in total, such that 1 should be executed before 2 and 3 should be executed before 4.

Thus, No. of possible sequences = $4! / 2! * 2! = 6$

Hence,

Possible sequences and values of a and b are as follows

1,2,3,4 = a = 4, b = 4

3,4,1,2 = a = 3, b = 3

1,3,2,4 = a = 4, b = 3

3,1,4,2 = a = 4, b = 3

1,3,4,2 = a = 4, b = 3

3,1,2,4 = a = 4, b = 3

Hence, option A is correct.

10. Solution:

required Sequence : BCA BCA BCA

Then T1 and T3 should wait until T2 print B

that means, even though thread T1 starts its execution prior then thread T2, it must be blocked therefore $S1 = 1, S2 = S3 = 0$.

11. Solution:

(b) 4 Page faults will happen

1(hit), 2 (hit), 4 (miss), 5 (miss), 2 (hit), 1(miss), 2(hit), 4(miss)

12. Solution:

Answer (b)

Explanation : Offset 1000 in segment ID 2 is crossing the segment ID 2 memory area. Here segment ID 2 memory area is 1527 to $1527 + 498 = 2025$.

13. Solution:

1. Maximum file size

$$\left(\frac{DB_{Size}}{DBA}\right)^3 \times DB_{Size}$$

2. Total size of file system

Total size of File System = (Disk Block size) $\times 2^{((DBA \text{ in bits}))}$

3. Maximum possible file size

$$\left(n_0 + n_1 \times \left(\frac{DB_{Size}}{DBA}\right) + n_2 \times \left(\frac{DB_{Size}}{DBA}\right)^2 + n_3 \times \left(\frac{DB_{Size}}{DBA}\right)^3\right) \times DB_{Size}$$

$$(2^7)^3 \times 2^{10} = 2^{31} = 2 \text{ GB}$$

14. Solution:

Since nothing is told about page tables, we can assume page table access time is included in memory access time.

So, average memory access time

$$= .9999 \times 1 + 0.0001 \times 10000$$

$$= 0.9999 + 1$$

$$= 1.9999 \text{ microseconds}$$

15. Solution:

If we have m resource's of same type and they are shared by 3 process's then

A needs 3 resource's of same type.

B needs 4 resource's of same type.

C needs 6 resource's of same type.

first of all we find out after how many resource's if we give them the system still be in Deadlock.

So

A needs 3 resource's of same type. If we only give 2 resource to A then it will still be in deadlock.

B needs 4 resource's of same type. If we only given 3 resource to B then it will still be in deadlock.

C needs 6 resource's of same type. If we only given 5 resource to C then it will still be in deadlock.

We have given total of 10 resources still the system is in deadlock. to break this deadlock we give them one more.

So Answer : A (11 resources of same type).

16. Solution:

Process P2 can be starved as for its execution it needs the process P1 to execute the V(x) operation before & in the case when only P2 is present in the system it can be starved.

Process P1 will not starve as after P2 has executed its V(x) operation & P1 loops on P(x) then there is a fair possibility for both P1 & P2 to get the next access to the semaphore x.

So, answer is option (D)

17. sol:

• System address space (VA) = 32bits

Page size (PS) = 4KB

• Total pages used by a process (N) = 2000 pages

• Page table entry size (PTE) = 4B

It is given in the question that OS allocates a page (means single page) for the outer page directory upon process creation.

So, total number of bits needed to represent entries in outer page table = $\log(PS / PTE) = \log((2^{12}) / (2^2)) = 10 \text{ bits}$

And total number of bits needed to represent entries in inner page table $32 - 10 - 12 = 10 \text{ bits}$

For X:

In each inner page table we can store 2^{10} entries. So, minimum number of page tables (PT) needed to store 2000 pages will be $\lceil 2000 / 2^{10} \rceil = 2 \text{ PT}$

... $X = 2 + 1$ (for outer PT) = 3

For Y:

For occupying maximum pages in 2-level page table, we need to place atleast one PTE from 2000 pages in every page of the inner page table equating to 2^{10} PT

.. $Y = 2^{10} + 1$ (for outer PT) = 1025

The value of $X + Y = 1028$





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18. Solution:

Process each page using FIFO

0 → Page fault (stored)

1 → page fault (stored)

2 → page fault (stored)

3 → Page fault (0 removed, 3 stored)

0 → page fault (1 removed, 0 stored)

1 → Page fault (2 removed, 1 stored)

4 → Page fault (3 removed, 4 stored)

0 → No page fault (already in memory)

1 → No page fault (already in memory)

2 → Page fault (0 removed, 2 stored)

3 → Page fault (1 removed, 3 stored)

4 → Page fault (4 re-added, 2 removed)

Total Page Faults:

10 page faults

Correct Answer : (B) 10

19. Sol:

Step 1: Calculate the need matrix

Need = Max – Allocation

P1: (7 - 0, 5 - 1, 3 - 0) = (7, 4, 3)

P2: (3 - 2, 2 - 0, 2 - 0) = (1, 2, 2)

P3: (9 - 3, 0 - 0, 3 - 3) = (6, 0, 0)

Step 2: Apply Banker's Algorithm

Initial Available Resources : (3, 3, 2)

Check if P1 can execute:

Needs (7, 4, 3) - Not enough resources

Check if P2 can execute

Needs (1, 2, 2) - can be satisfied

Allocate P2's resources back: New Available = (5, 3, 2)

Check if P3 can execute:

Needs (6, 0, 0) Not enough resources

Check if P1 can execute again

Needs (7, 4, 3) Still not enough

Since no further process can execute, the system is in deadlock.

Correct Answer: (B) No, deadlock.

20. Solution:

1. Sequence Based on shortest seek time first (SSTF)

50 → 43 → 24 → 16 → 82 → 140 → 170 → 190

2. Total Head Movement

$|50-43| + |43-24| + |24-16| + |16-82| + |140-170| + |170-190|$
 $= 7 + 19 + 8 + 66 + 58 + 30 + 20$
 $= 208$

Correct Answer (A) 208 cylinders

